PS 2010: 5. Random Variable 1

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- You all did very well for the assignment 1 and 2
- Middle term exam next week
- All you need to know is in slides and homework assignment
- Course outline changes. See you syllabus
- Introduction to Probability
- Foundations of Statistics with R
- Readings for next few weeks

Agenda

- Expectation and Variance
- Distribution of discrete random variable
- Distribution of continuous random variable

Bayes Rule Revisit

It is estimated that 50% of emails are spam emails. Some software has been applied to filter these spam emails before they reach your inbox. A certain brand of software claims that it can detect 99% of spam emails, and the probability for a false positive (a non-spam email detected as spam) is 5%.

Now if an email is detected as spam, then what is the probability that it is in fact a non-spam email?

Bayes Rule Revisit

- You were hired to advise a political candidate, a Republican, running for office in a two-candidate race against a Democrat. You were unfamiliar with the district so your (uniformed) prior belief about the likelihood your candidate (the Republican) would win was a coin flip (50%).
- However, a poll (conducted once every election) has just been released, and it shows that your candidate is going to lose.
- You know that this poll predicts Democratic victories 40% of the time and that it has a false negative rate of 25%. False negative, in this instance, means that in 25% of the cases in which the Republican wins, the poll predicts the opposite, a Democratic victory.
- Employ Bayes' Law to update your belief about your candidate's chances of winning the election? What is your updated probability of victory (your posterior belief)?

Expectation

- Y is a discrete random variable, then the expected value of Y denoted as $E(Y)$ is $\mu = E(y) = \sum yf(y)$
- If Y is a continuous variable, then $\mu = E(y) = \int_{-\infty}^{\infty} yf(y)$
- Expectation represent the mean of random variables
- Let Y be a random variable with pdf f(y), the expected value of the discrete random variable g(Y) is

$$\mu_{g(Y)} = E[g(Y)] = \sum_y g(y)f(y)$$

- The expected value of the continuous random variable is

$$\mu_{g(Y)} = E[g(Y)] = \int_{-\infty}^{\infty} g(y)f(y)dy$$

Example

A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

Example

Let X be the random variable that denotes the life in hours of a certain electronic device. The probability density function is

$$f(x) = \begin{cases} \frac{20{,}000}{x^3}, & x > 100 \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected life of this type of device.

Example

The length of time required by students to complete a one-hour exam is a random variable with a density function given by

$$f(x) = \begin{cases} cx^2 + x, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

- Find the value c for which f(x) is a valid density function
- Find F(x)
- Find $P(x \leq 0.5)$
- Find the expected length.
- Find $g(x) = 4X + 3$

Example

Suppose that the number of cars X that pass through a car wash between 4 : 00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

x	4	5	6	7	8	9
$P(X = x)$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$

Let $g(X) = 2X - 1$ represent the amount of money, in dollars, paid to the attendant by the manager. Find the attendant's expected earnings for this particular time period.

Rules of Expectation

Let X and Y be random variables, and a and b be arbitrary constants. Then

- $E(a) = a$
- $E(aX) = aE(X)$
- $E(aX + b) = aE(X) + b$
- $E(aX + bY) = aE(X) + bE(Y)$
- If X and Y are independent, then $E(XY) = E(X)E(Y)$, but generally, $E(XY) \neq E(X)E(Y)$

Example

A busy student must complete 3 problem sets before doing laundry. Each problem set requires 1 day with probability 2/3 and 2 days with probability 1/3. Let B be the number of days a busy student delays laundry. What is $E(B)$?

Variance

Definition: If X is a random variable with mean $E(X) = \mu$, the variance of a random variable X is defined to be the expected value of $(X - \mu)^2$:

$$V(X) = E[(X - \mu)^2]$$

The square root of X is called the standard deviation

Example: The probability distribution for a random variable Y is given in the table below. Find the mean, variance, and standard deviation of Y.

y	$p(y)$
0	1/8
1	1/4
2	3/8
3	1/4

Variance

$$
\begin{aligned}
V(X) &= E[(X - E(X))^2] \\
&= E[X^2 - 2X * E(X) + \{E(X)\}^2] \\
&= E(X^2) - 2E(X)E(X) + \{E(X)\}^2 \\
&= E(X^2) - 2\{E(X)\}^2 + \{E(X)\}^2 \\
&= E(X^2) - \{E(X)\}^2
\end{aligned}
$$

where we call $E(X)$ as the first moment, and $E(X^2)$ as the second moment

Example: A pdf $f(x) = \frac{1}{b-a}$ at interval [a, b], calculate the variance of random variable X

Rules of Variance

Let X and Y be random variables, and a and b be arbitrary constants. Then

- $V(a) = 0$
- $V(aX) = a^2 V(X)$
- $V(X + b) = V(X)$
- $V(aX + b) = a^2 V(X)$
- If X and Y are independent, $V(X + Y) = V(X) + V(Y)$

Distribution

- Distribution for discrete random variable
  - Bernoulli and Binomial
  - Negative Binomial
  - Poisson
- Distribution for continuous random variable
  - Uniform
  - Normal
  - Chi-square
  - Student t

Bernoulli and Binomial

- A random variable that takes two distinct values is called a **Bernoulli Random Variable**
- Success equals to 1 with probability p, and fail 0 with 1-p
- Example: flip a coin, election, classification
- **Binomial Random Variable** is the sum of n independently and identically distributed Bernoulli random variables.
- The probability distribution of this random variable is called binomial distribution.
- Example: conflict early warning system for the next ten times

Binomial

The probability mass function (PMF) of a binomial random variable with success probability p and n trials is given by

$$f(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}.$$

The cumulative distribution function (CDF) can be written as

$$F(x) = P(X \leq x) = \sum_{k=0}^{x} \binom{n}{k} p^k (1-p)^{n-k},$$

for $x = 0, 1, \ldots, n$.

Example

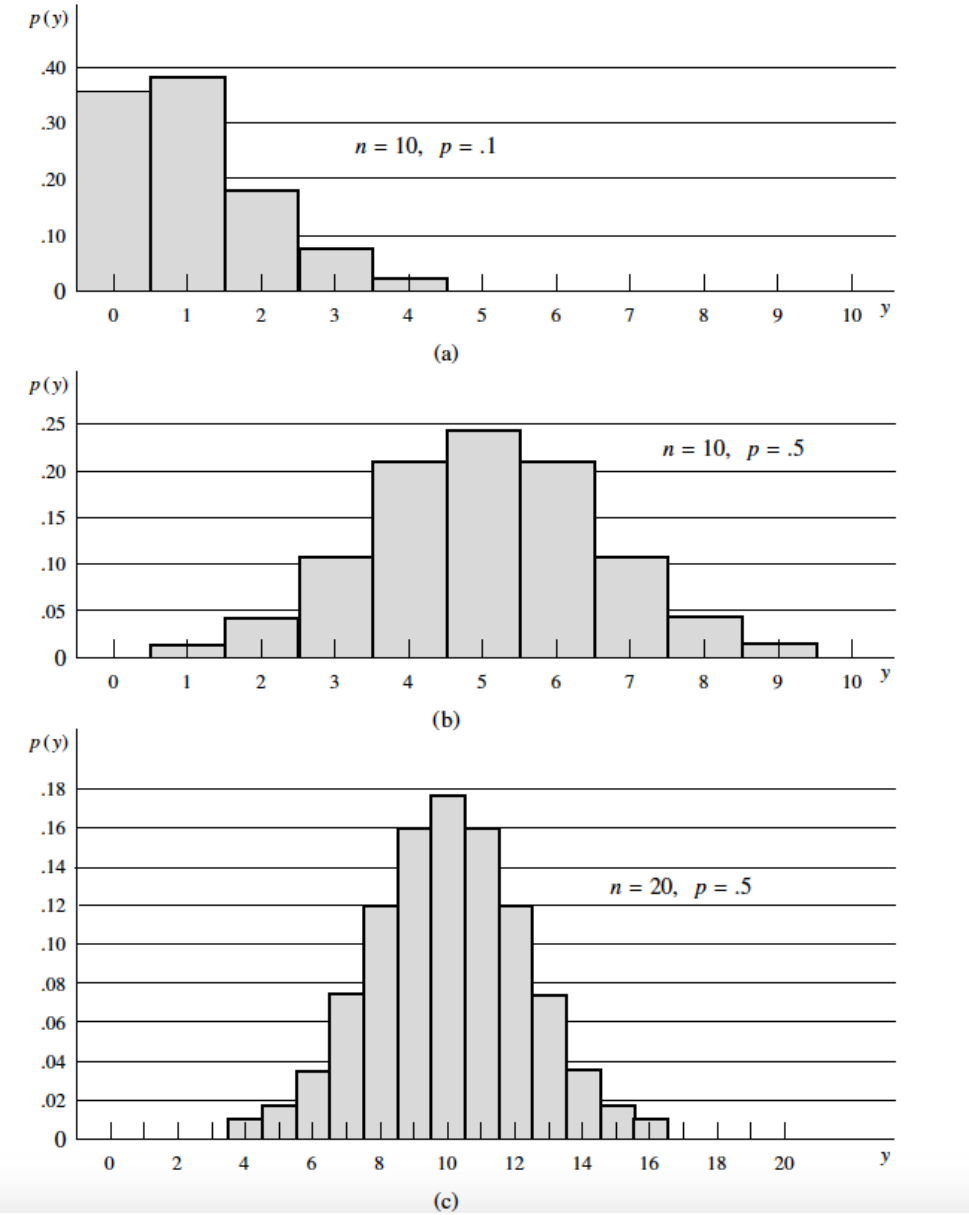
- The probability that a certain kind of component will survive a shock test is 3/4. Find the probability that exactly 2 of the next 4 components tested survive.
- 49% of Americans likes Trump, if we randomly select 10 people from public, what is the probability of at least 9 people like Trump
- In order to verify the accuracy of their financial accounts, companies use auditors on a regular basis to verify accounting entries. The company's employees make erroneous entries 5% of the time. Suppose that an auditor randomly checks three entries. Find the probability distribution for Y , the number of errors detected by the auditor.

Mean and Variance of Binomial

Let X be a binomial random variable based on n trials and success probability p, then

$$\mu = E(X) = np \text{ and } \sigma^2 = V(X) = np(1-p)$$

Binomial Distribution



Negative Binomial

- Negative binomial also count events, but this time we care about the fixed number of success occur
- The number of X trials required to produce k successes is a negative binomial random variable
- A binomial random variable X has the following pdf:

$$f(x) = P(X = x) = \begin{pmatrix} x-1 \\ k-1 \end{pmatrix} p^k (1-p)^{x-k}.$$

- And the mean $\mu = E(X) = \frac{K}{p}$, and $\sigma^2 = V(X) = \frac{k(1-p)}{p^2}$

Example

- A conflict early warning system can predict successfully 55% percent of potential conflicts in the experimental stage. Now it handed over to another team for testing, and it can pass the testing if the system can make 4 success predictions in 7 testings.
- What is the probability that system pass the test at the 6th test
- What is the probability that the system pass the test

Example

Qing can not decide what to drink on Friday night, so he decides to repeatedly random draw from bottles of Scotch, Bourbon, and Vodka, and take shots. Qing know that 60% of them is Scotch, 30% of them is Bourbon, and 10% of them is Vodka.

- Find the probability that Qing's fifth Scotch shot happens when he draw eight times
- Find the probability that Qing drinks 5 Scotch if he draws 8 times

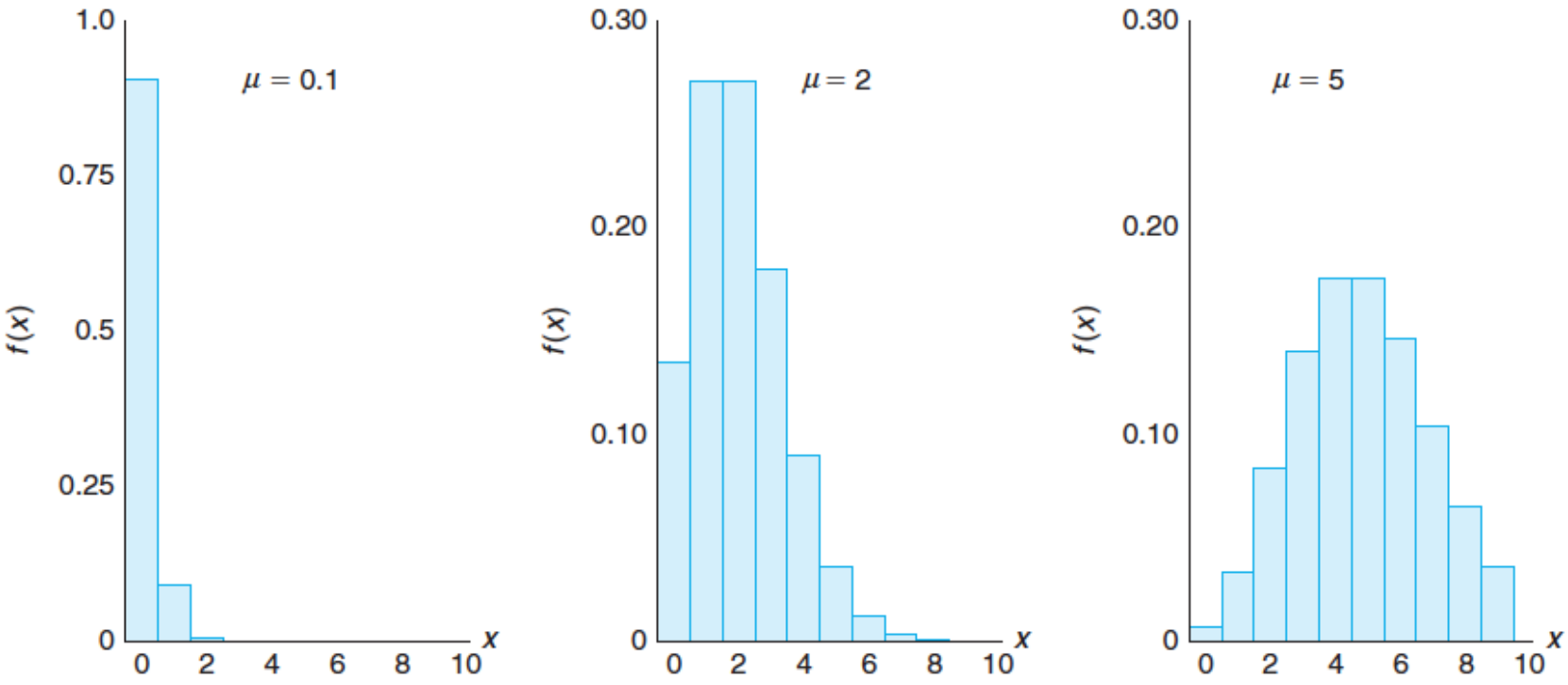
Poisson Distribution

- The number X of events occurring in one time interval is called a Poisson random variable
- Example: conflicts happening in one year, # of people dead due to Covid in a day
- The Poisson distribution for random variable X with mean λ, has the following pdf:

$$f(x) = p(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

- The means of Poisson distribution is $\mu = \lambda$ and $V(x) = \lambda$

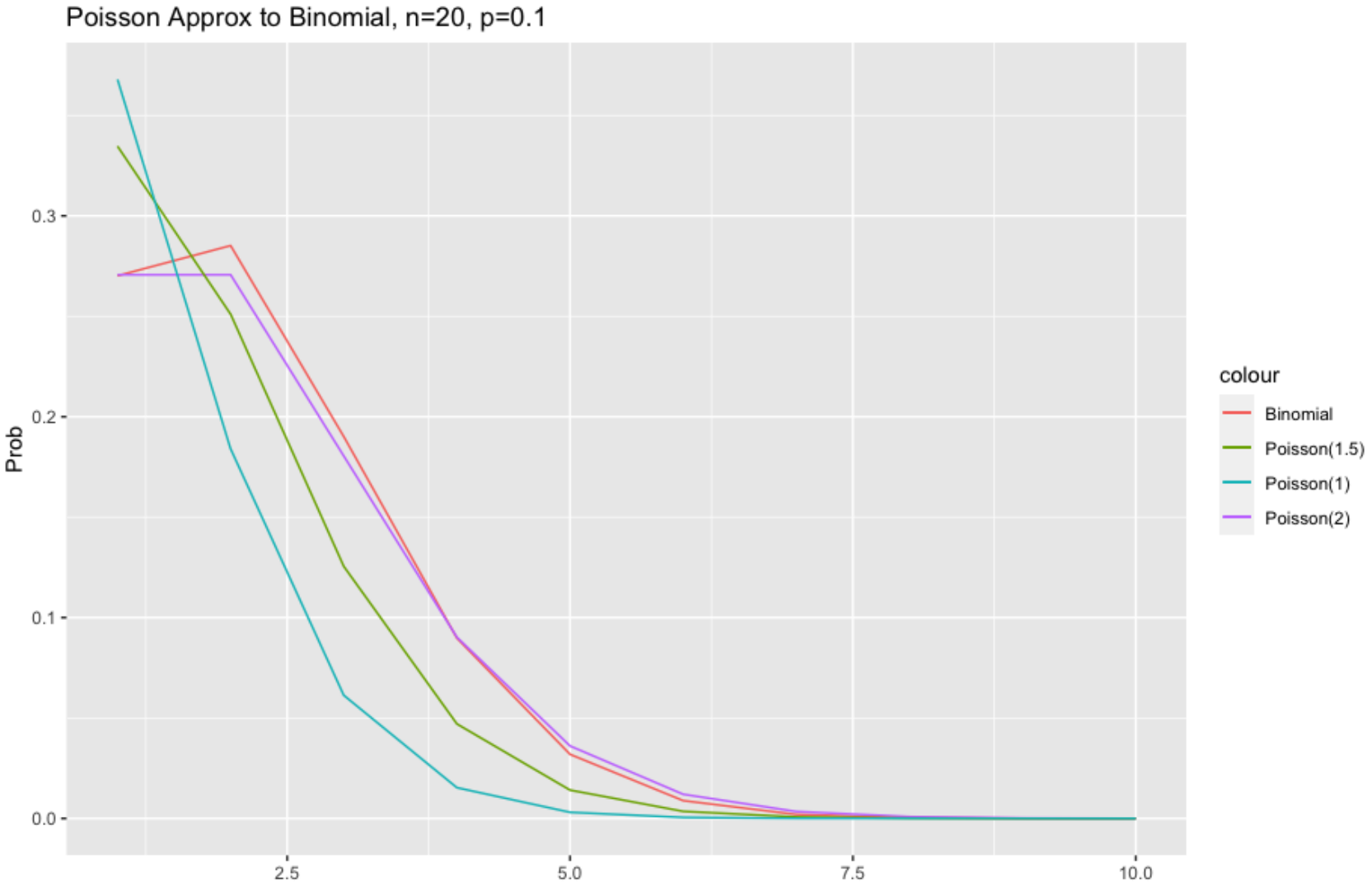
Poisson Example



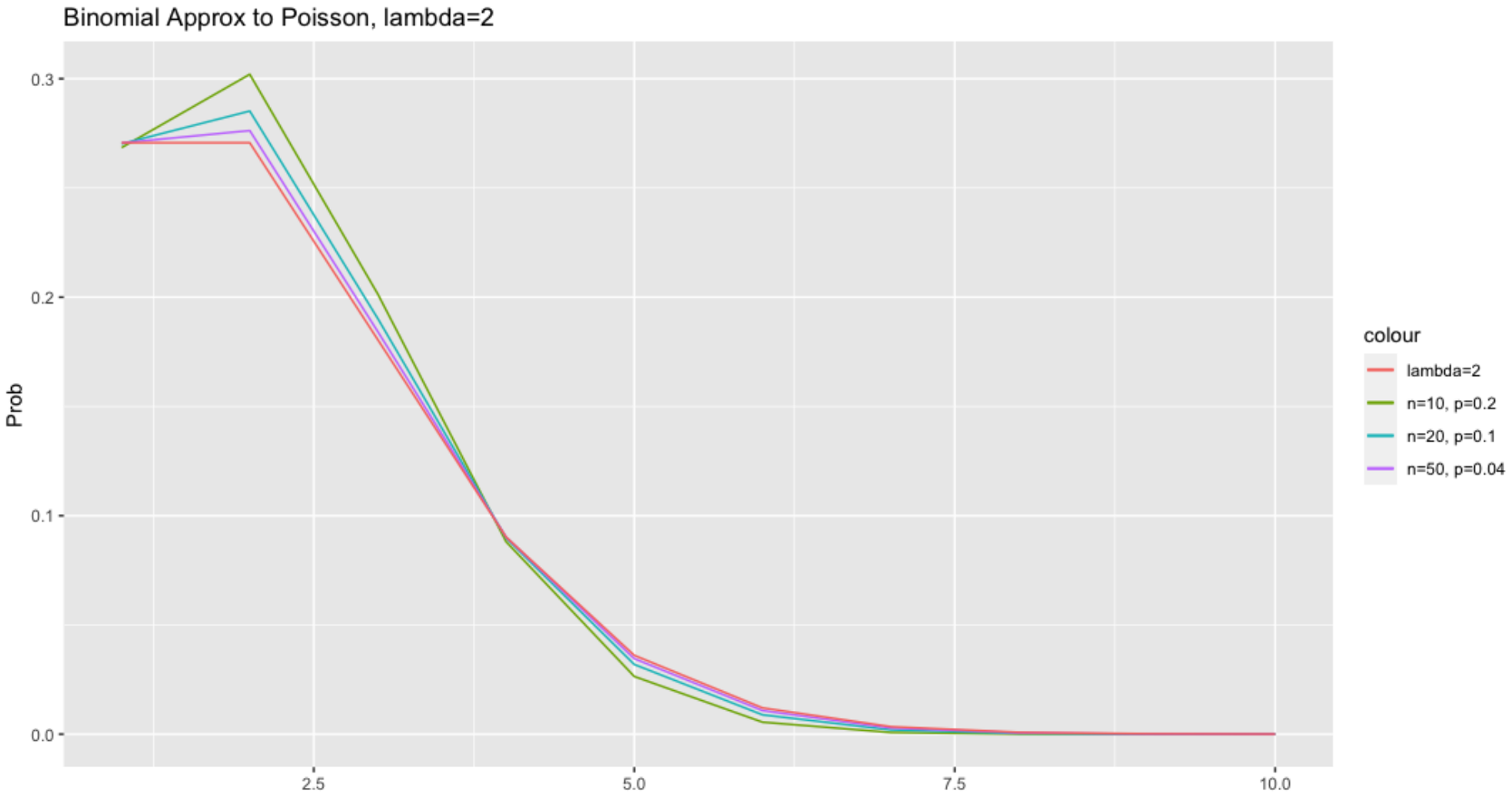
Example

- The average conflict of a year is 5, what is the probability that 6 conflicts happened in a year?
- The average number of bills passed by congress is 36 a year, what is the probability that 45 bills passed in a year?

Approximation of Binomial Distribution by a Poisson Distribution



Approximation of Poisson Distribution by a Binomial Distribution



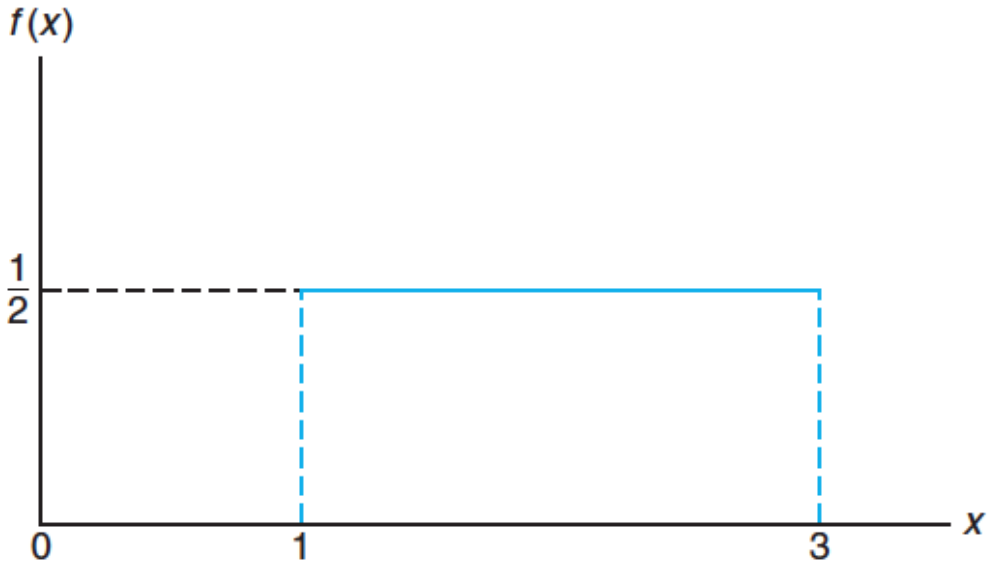
Uniform Distribution

- A very simple continuous probability distribution
- Pdf for a uniform random variable X on the interval [a,b]:

$$f(x) = \begin{cases} \frac{1}{B-A}, & A \leq x \leq B \\ 0, & \text{elsewhere} \end{cases}$$

- The density function forms a rectangle with base B-A and constant height $\frac{1}{B-A}$
- The mean of uniform random variable is $\mu = \frac{A+B}{2}$ and $\sigma^2 = \frac{(B-A)^2}{12}$

Uniform Distribution



Example

Hillman library require students to reserve study room for no more than 4 hours per person. In fact, it can be assumed that the length X of a reservation has a uniform distribution on the interval [0, 4].

- What is the probability density function?
- What is the probability that any given conference lasts at least 3 hours?

Normal Distribution

- Normal or Gaussian distribution takes the value from $-\infty$ to ∞
- The density function of a normal random variable X, with mean μ and variance σ^2 is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \ -\infty \leq x \leq \infty$$

- There is no analytical tractable form of cdf for a normal distribution.
- However, you can always calculate the probability between interval [a,b] as the area under the normal distribution curve

Standard Normal

- X is normal random variable with mean μ and standard deviation σ, denoted as $N(\mu, \sigma^2)$
- If a normal distribution has mean 0 and variance 1, then it is called **standard normal distribution**
- If c is a constant number, then $\mathrm{Z} = \mathrm{X} + \mathrm{c}$ is also a normal distribution with $\mathrm{Z} \sim N(\mu + c, \sigma^2)$
- If $\mathrm{Z} = \mathrm{cX}$, then $\mathrm{Z} \sim N(c\mu, (c\sigma)^2)$
- $z - score = \frac{X-\mu}{\sigma} \sim N(0, 1)$

Standard Normal

- For a standard normal, 1 standard deviation from the mean covers 68% of the observations

$$P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68$$

- 2 standard deviations from the mean covers 95%

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

- z-score locates a point by how many in units of the standard deviation from the mean.
  Very useful when construct confidence interval and hypothesis testing
- A simple way to do standardization for variables

Example

- Given that X has a normal distribution with $\mu = 300$ and $\sigma = 50$, find the probability that X assumes a value greater than 362.
- Suppose the final grade for this class is normally distributed with $\mu = 75$ and $\sigma = 10$, what fraction of the scores lies between 80 and 90?
- If your instructor want 90% of students get an A and the rest of them are Bs, how should the instructor set the cutoff point?

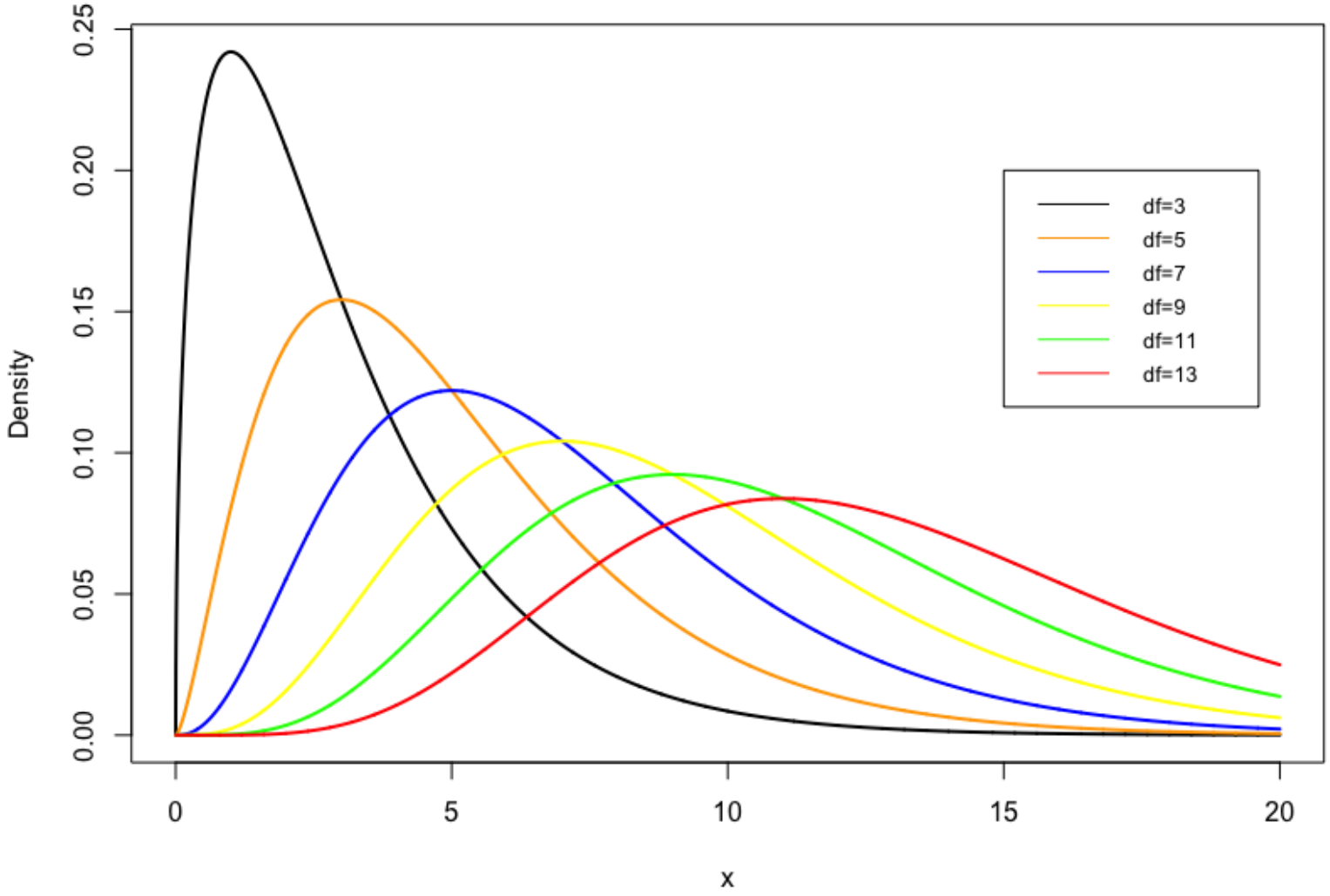
Chi-square

Several other commonly seen distributions:

- Chi-square, χ^2, distribution: If $Z_1, \ldots, Z_k$ are independent standard normal random variables, then

$$X^2 = Z_1^2 + \cdots + Z_k^2$$

  has a χ^2 distribution with k degrees of freedom.
- χ^2 distribution converges to normal distribution
- Useful for hypothesis testing for multivariate

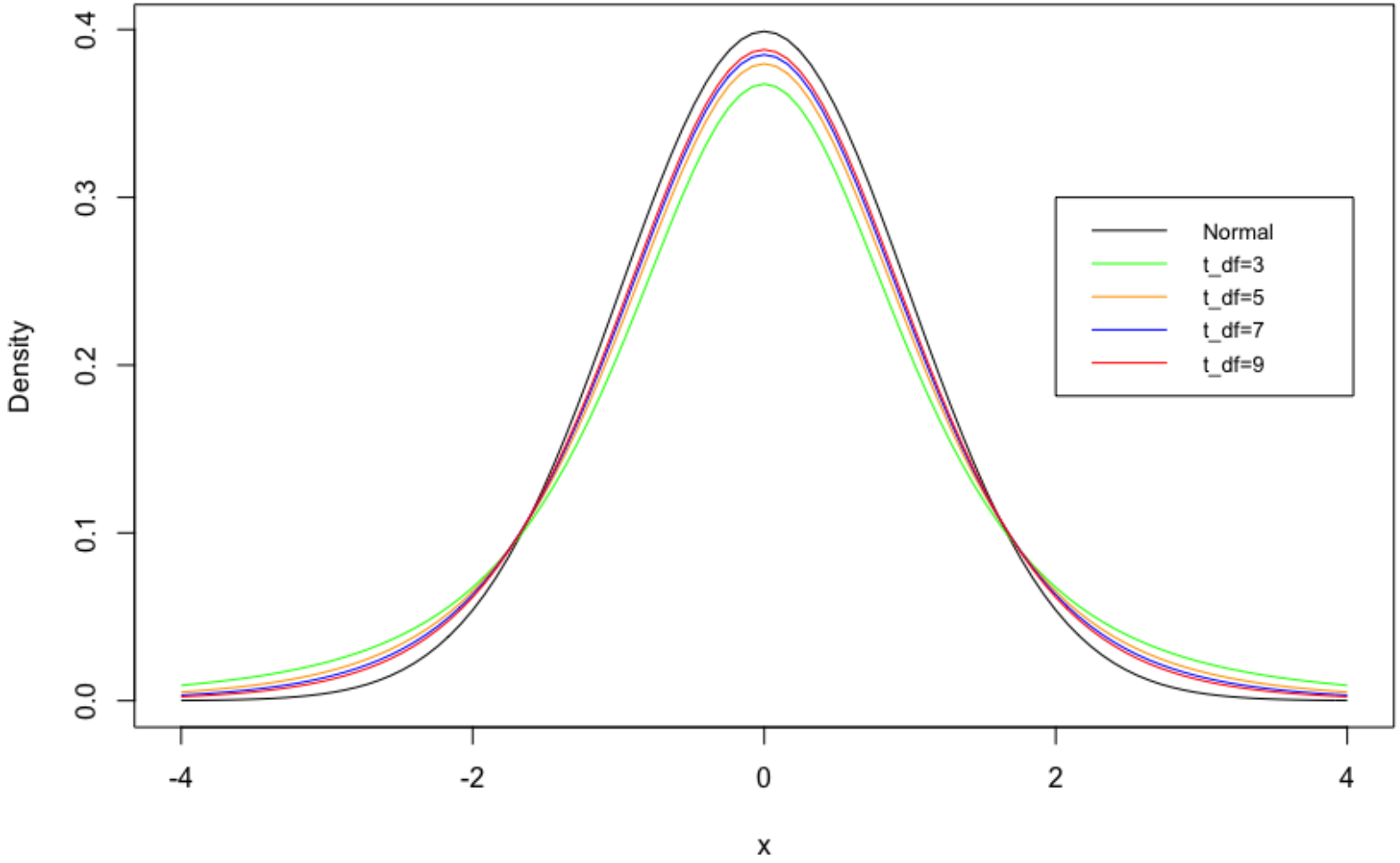

Density
0.00
0.05
0.10
0.15
0.20
0.25
0
5
10
15
20
x
df=3
df=5
df=7
df=9
df=11
df=13

Student t

If Z is a standard normal random variable and if X^2 is a χ^2 random variable with k degrees of freedom, then

$$T = \frac{Z}{\sqrt{X^2/k}}$$

has a t distribution with k degrees of freedom.

- t densities are symmetric, bell-shaped, and centered at 0 just like the standard normal density, but are more spread out (higher variance).
- As the degrees of freedom increases, the t distributions converge to the standard normal.
- t distributions will be useful for statistical inference for one or more populations of quantitative variables.

t distribution